ARCQuant: Boosting NVFP4 Quantization with Augmented Residual Channels for LLMs

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Problem

Activation outliers still hurt W4A4 NVFP4, even with fine-grained block scaling.

ARC

AUGMENTED RESIDUAL CHANNELS

Augmented Residual Channels: quantize outlier residuals and append them as extra reduction channels.

Result

W4A8-level accuracy while preserving one unified NVFP4 GEMM path.

Why Existing PTQ Breaks

The bottleneck is not just numerical precision; it is preserving NVFP4's hardware-friendly structure.

Rotation
Hadamard rotation spreads channel outliers across blocks, reducing peak error but breaking NVFP4's local block-wise isolation.

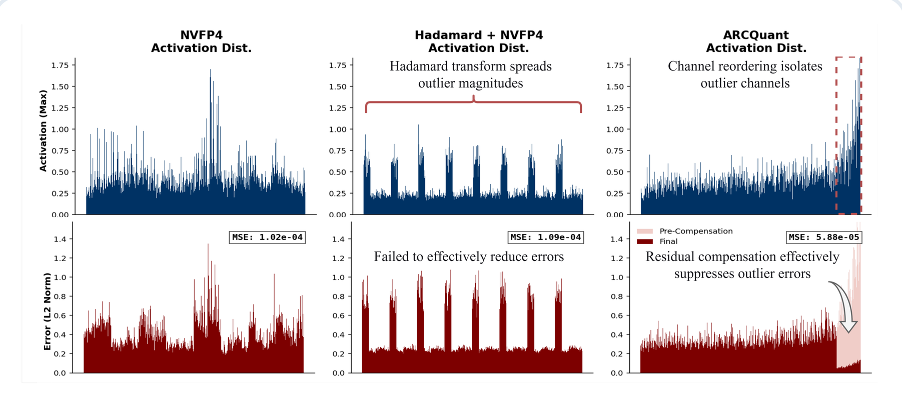
Smoothing
Activation smoothing shifts scale between X and W, yet severe 4-bit clipping leaves residual outlier energy that still dominates error.

Mixed Precision
Higher-precision outlier branches recover accuracy, but introduce separate data paths and prevent one clean Tensor Core GEMM.

Core Idea

ARCQuant keeps activation outliers localized, then converts their quantization residuals into extra K-dimension channels instead of a separate compute branch.

$$Y \approx Q(X)Q(W)^T + Q(R_o)Q(W_o)^T$$
$$= Q([X \mid R_o])Q([W \mid W_o])^T$$



ARC isolates outliers and suppresses residual error; Hadamard spreads outlier magnitude across blocks.

The main activation and residual correction stay in the same NVFP4 format, so GEMM sees one augmented matrix multiply.

Contributions

Accuracy
Best W4A4 results across LLaMA and Qwen settings.

Analyses
Error analyses show dual-stage NVFP4 has an error bound comparable to MXFP8.

Systems
Fused quantization plus standard optimized CUTLASS GEMM.

Deployment
Validated on RTX 5090 and RTX PRO 6000 GPUs.

Paper & Code

TensorRT-LLM Support
ARCQuant has been merged upstream as 2FP4 / ARCQuant support.

Paper

Code

Contact **actypedef@gmail.com**

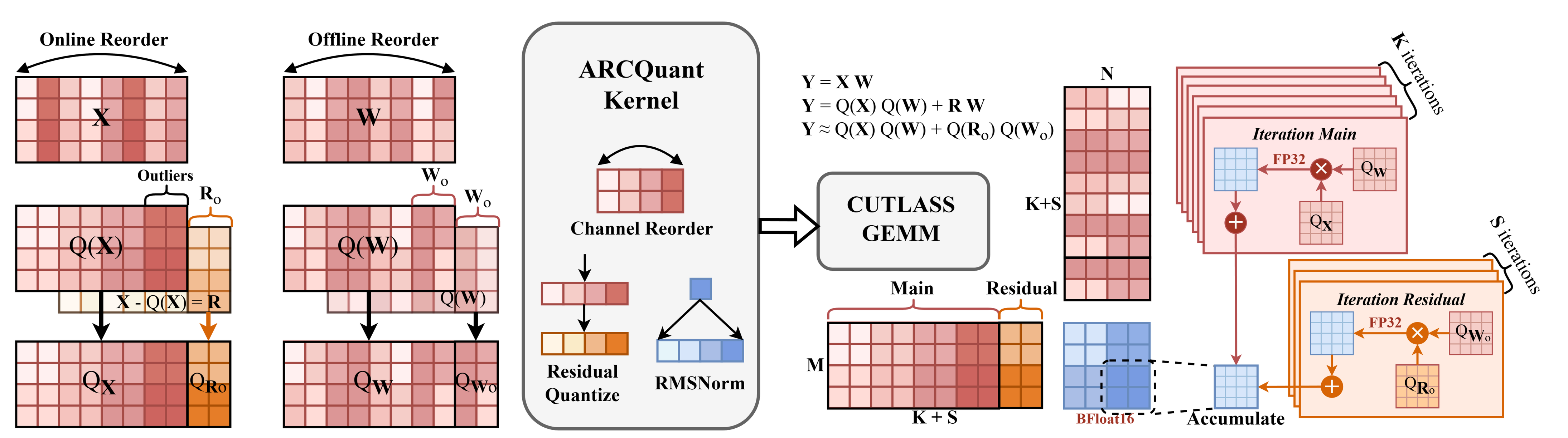
Method: Augmented Residual Channels

1 Calibrate
Compute reorder indices and choose each layer's augmented size S.

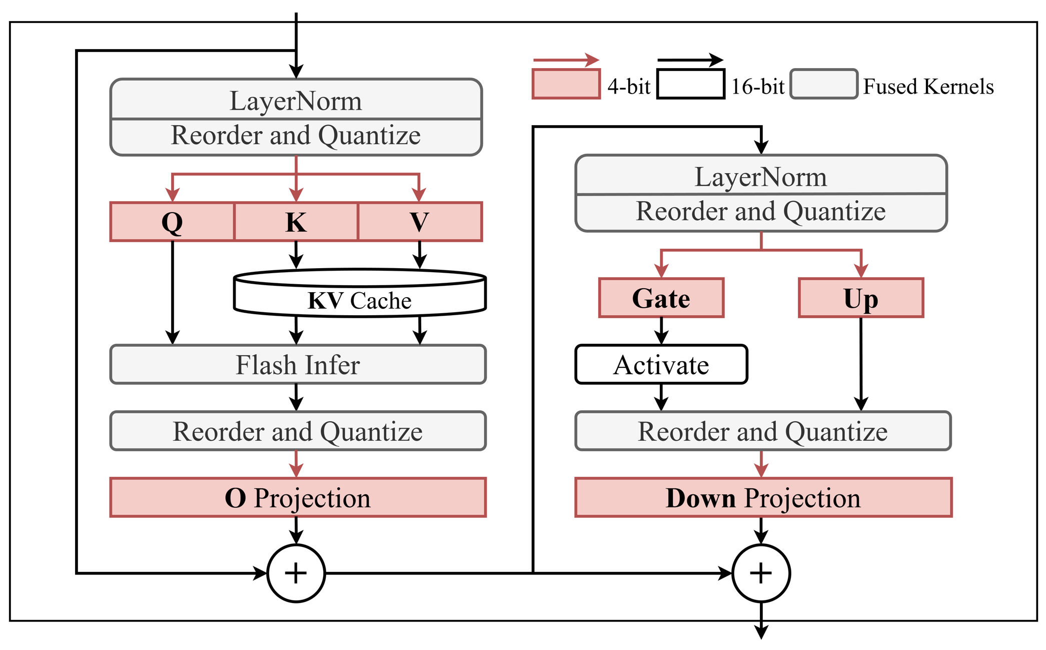
2 Weight Path
Reorder W, quantize it, then duplicate and append selected outlier columns.

3 Fused X Path
Reorder X, quantize primary values, quantize residuals, then concatenate [X | R_o].

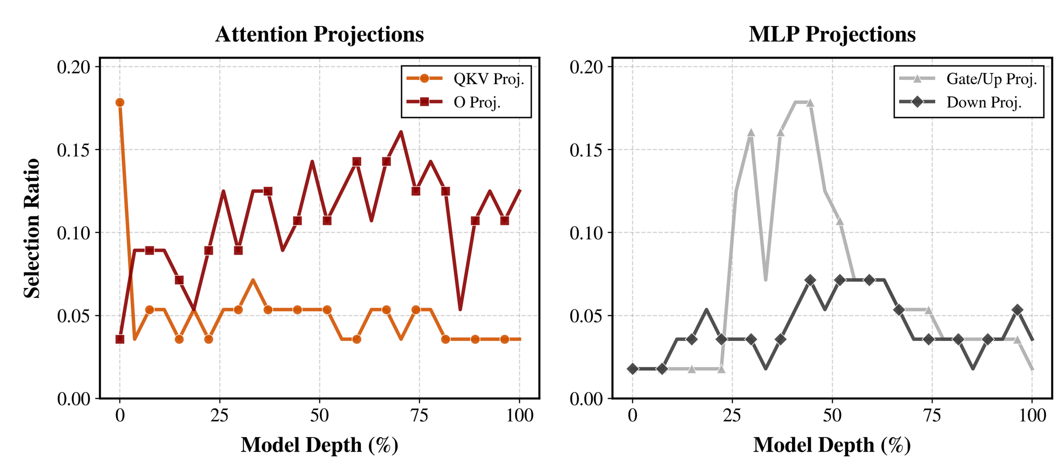
4 Standard GEMM
Run one standard NVFP4 GEMM over K_{in} + S to accumulate main and residual terms.



Main and residual accumulation are fused into an extended reduction dimension.



Transformer block integration.

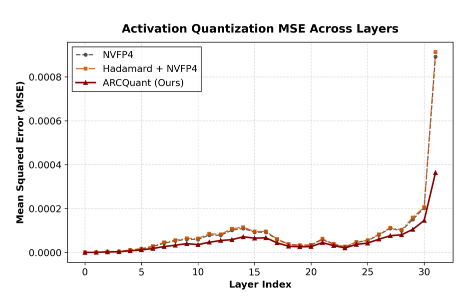


Layer-wise selected residual channels S.

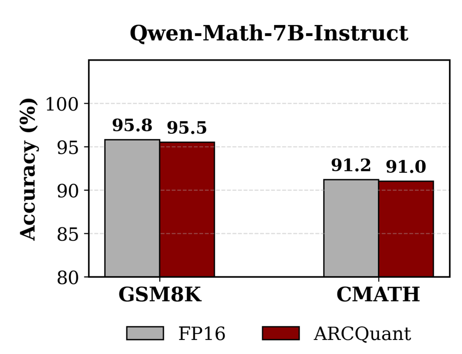
Accuracy

Model	BF16	W4A8 MXFP	ARCQuant	BF16 Ret.
Llama 3.1-8B	72.56	70.59	70.90	97.7%
Qwen2.5-7B	70.97	69.15	70.28	99.0%
Qwen2.5-32B	74.82	75.00	74.80	100.0%

Average 0-shot accuracy. Retention is ARCQuant / BF16.



ARCQuant consistently suppresses NVFP4 activation MSE.



Qwen2.5-Math-7B-Instruct on GSM8K and CMATH.

Representative Extended Results

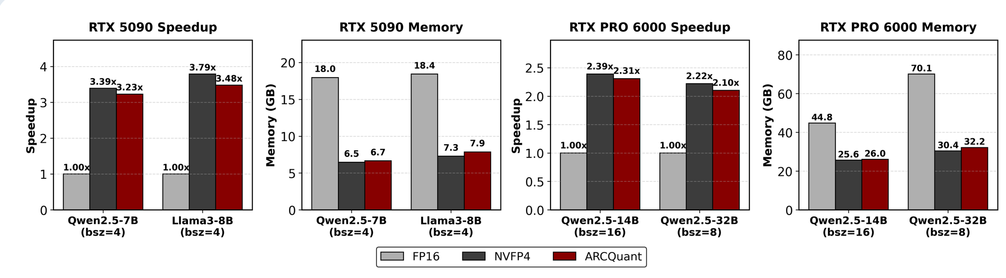
76.08 / 76.61
Llama 3.1-70B
Avg. 0-shot / MMLU. BF16: 78.20 / 78.57; improves over NVFP4 RTN by +0.65 / +0.50.

78.03 / 68.71
Mixtral 8x7B
Avg. 0-shot / MMLU. BF16: 78.46 / 70.30; improves RTN avg. 0-shot by +0.40.

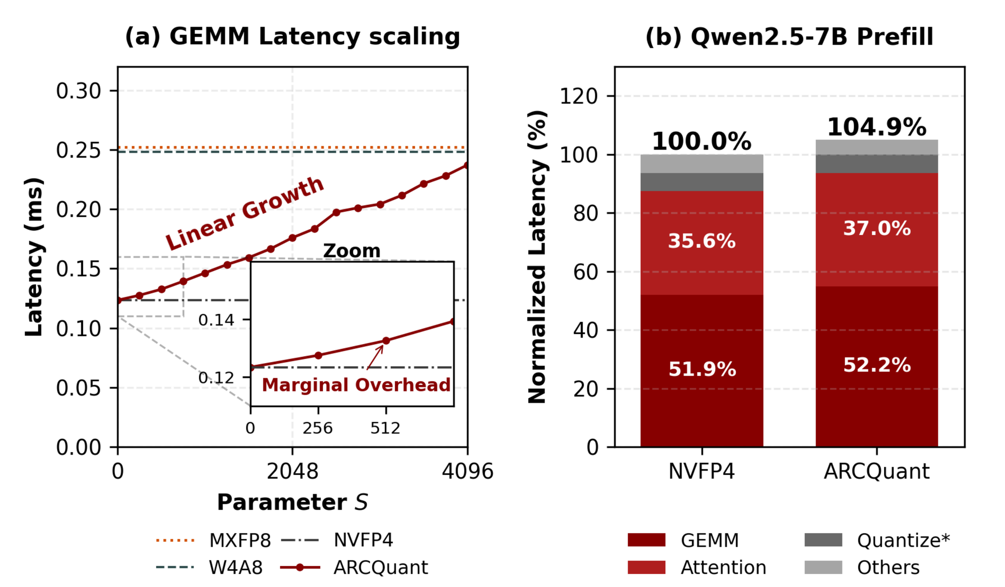
86.0 / 68.3
Qwen2.5-Coder-7B
HumanEval / MBPP+ pass@1. BF16: 84.1 / 67.2.

95.5 / 91.0
Qwen2.5-Math-7B
GSM8K / CMATH accuracy. BF16: 95.8 / 91.2.

Efficiency



Prefill speed and memory on RTX 5090 and RTX PRO 6000.



Residual overhead stays modest in the typical S range.

3×
speedup over FP16 in practical prefill settings

2.08×
decode speedup over FP16 at sequence length 2048

3-9%
latency overhead over uncompensated NVFP4

4.9%
total prefill overhead in Qwen2.5-7B breakdown